

Section32: SQL

Documentation: <https://www.w3schools.com/sql/>

241. SQL Commands: CREATE Table and INSERT Data

- Main things we do in SQL (CRUD):
 - Create
 - Read
 - Update
 - Destroy
- Create:

```
CREATE TABLE products (  
  id INT NOT NULL,  
  name SQL,  
  price MONEY,  
  PRIMARY KEY (id)  
);
```

- Insert:
 - No need to mention column names if insertion is done on all columns.

```
INSERT INTO products  
VALUES (1, 'Pen', 1.20);
```

- Insert while skipping fields

```
INSERT INTO products (id, name)  
VALUES (2, 'Pencil');
```

242. SQL Commands: READ, SELECT and WHERE

- Selecting all columns:
 - SELECT * FROM 'products';
- Select particular columns:
 - SELECT name, price FROM 'products';
- Select columns based on a condition:
 - SELECT * FROM 'products' WHERE id=1;

243. Updating Single Values and Adding Columns in SQL:

- Updating particular row in a column:
 - UPDATE products
SET price = 0.08
WHERE id = 2;
- Change table:
 - ALTER TABLE products
ADD stock int;

244. SQL Commands: DELETE

- Deleting a particular row in a table
 - DELETE FROM products
WHERE id = 1;

245. Understanding SQL Relationships, Foreign Keys and Inner Joins

- Foreign key:

```
CREATE TABLE Orders (  
    OrderID int NOT NULL,  
    OrderNumber int NOT NULL,  
    PersonID int,  
    PRIMARY KEY (OrderID),  
    FOREIGN KEY (PersonID) REFERENCES Persons(PersonID)  
);
```

```
ALTER TABLE Orders  
ADD FOREIGN KEY (PersonID) REFERENCES Persons(PersonID);
```

- Join

```
SELECT  
    Orders.OrderID,  
    Customers.first_name,  
    Customers.last_name,  
    Customers.address  
FROM Orders  
INNER JOIN Customers ON Orders.customer_id = Customers.id;
```